Practice Set A

Class 12 Mathematics, Chapter 9: Differential Equations

Maximum marks: **51** **20** questions

Name _____ Date _____ Score _____ / 51

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ, CBSE 2023 — 1 mark

The order and the degree of the differential equation $\left(1+3\frac{dy}{dx}\right)^2 = 4\frac{d^3y}{dx^3}$ respectively are:

(A) $1,\ \frac{2}{3}$ **(B)** $3,\ 1$ **(C)** $3,\ 3$ **(D)** $1,\ 2$

2 MCQ, CBSE 2023 — 1 mark

The degree of the differential equation $\sin x + \cos\left(\frac{dy}{dx}\right) = y^2$ is:

(A) 2 **(B)** 1 **(C)** not defined **(D)** 0

3 MCQ, CBSE 2023 — 1 mark

The integrating factor of the differential equation $(1-y^2)\frac{dx}{dy} + yx = ay,\ (-1 < y < 1)$ is:

(A) $\frac{1}{y^2-1}$ **(B)** $\frac{1}{\sqrt{y^2-1}}$ **(C)** $\frac{1}{1-y^2}$ **(D)** $\frac{1}{\sqrt{1-y^2}}$

4 MCQ, NCERT Ex 9.2 Q12 — 1 mark

The number of arbitrary constants in the particular solution of a differential equation of third order are:

(A) 3 **(B)** 2 **(C)** 1 **(D)** 0

5 MCQ, CBSE SQP 2024-25 — 1 mark

What is the general solution of the differential equation $e^{y'} = x$?

(A) $y = x\log x + c$ **(B)** $y = x\log x - x + c$

(C) $y = x\log x + x + c$ **(D)** $y = x + c$

Q No.	Question	Marks
6	Assertion-Reason, Board-pattern **Assertion (A):** $y = ae^{3x} + be^{-2x}$ is a solution of the differential equation $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 6y = 0$. **Reason (R):** The number of arbitrary constants in the general solution of a differential equation is equal to its order. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer, PYQ-style Find the general solution of the differential equation $\frac{dy}{dx} = e^{x-y} + x^2e^{-y}$.	2 marks

Level 2 — 18 marks

8	Short answer, NCERT Ex 9.3 Q20 In a bank, principal increases continuously at the rate of $r\%$ per year. Find the value of r if ₹100 double itself in 10 years $(\log_e 2 = 0.6931)$.	2 marks
9	Short answer, NCERT Misc. Ex. Q7 Find the particular solution of the differential equation $(1 + e^{2x})\,dy + (1 + y^2)\,e^x\,dx = 0$, given that $y = 1$ when $x = 0$.	3 marks
10	Short answer, PYQ-style Find the general solution of the differential equation $x\frac{dy}{dx} + y = x\cos x + \sin x$, $(x \neq 0)$.	3 marks
11	Short answer, CBSE 2023 Solve the differential equation $(x^2 - y^2)\,dx + 2xy\,dy = 0$.	3 marks
12	Short answer, PYQ-style Verify that $y = A\cos(\log x) + B\sin(\log x)$ is a solution of the differential equation $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = 0$.	2 marks
13	Short answer, Board-pattern Find the general solution of the differential equation $\frac{dy}{dx} = x\log x$, $(x > 0)$.	2 marks
14	Short answer, Board-pattern Find the particular solution of the differential equation $\frac{dy}{dx} + y = \cos x - \sin x$, given that $y = 3$ when $x = 0$.	3 marks

Q No.	Question	Marks
Level 3		**25 marks**
15	Long answer, NCERT Ex 9.4 Q1 Show that the differential equation $(x^2 + xy)\,dy = (x^2 + y^2)\,dx$ is homogeneous and solve it.	5 marks
16	Short answer, CBSE SQP 2026-27 Find the particular solution of the differential equation $x\,dy + (y - x^2 e^x)\,dx = 0$, given that $y(1) = 0$.	3 marks
17	Long answer, CBSE SQP 2025-26 Find the particular solution of the differential equation $2y\,e^{x/y}\,dx + \left(y - 2x\,e^{x/y}\right)dy = 0$, given that $y(0) = 1$.	5 marks
18	Short answer, CBSE 2023 Find the general solution of the differential equation $(xy - x^2)\,dy = y^2\,dx$.	3 marks
19	Long answer, PYQ-style Find the particular solution of the differential equation $y\,e^y\,dx = \left(y^3 + 2x\,e^y\right)dy$, given that $x = 0$ when $y = 1$.	5 marks
20	Case study, Board-pattern **Case Study:** The population of a village increases continuously at a rate proportional to the number of its inhabitants present at any time. The population was 20,000 in the year 2019 and 25,000 in the year 2024. Measure the time t in years from 2019. **(i)** Write the differential equation for the population P at time t and state its order and degree. — 1 mark **(ii)** Solve it to express P in terms of t and find the constant k. — 1 mark **(iii)** In which year will the population be 31,250? — 2 marks	4 marks

Answer key

Chapter 9: Differential Equations, Practice Set A. Full step-wise solutions are in the solutions sheet.

Level 1

1	(B) $3,\ 1$
2	(C) not defined
3	(D) $\dfrac{1}{\sqrt{1-y^2}}$
4	(D) 0
5	(B) $y = x\log x - x + c$
6	(B) Both A and R are true, but R is not the correct explanation of A.
7	$e^y = e^x + \dfrac{x^3}{3} + C$

Level 2

8	$r = 6.931$, i.e. the rate is 6.931% per year
9	$\tan^{-1} y + \tan^{-1}(e^x) = \dfrac{\pi}{2}$, i.e. $y = e^{-x}$
10	$xy = x\sin x + C$, i.e. $y = \sin x + \dfrac{C}{x}$
11	$x^2 + y^2 = Cx$
12	$x^2\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} + y = 0$ holds, so the function is a solution.
13	$y = \dfrac{x^2}{2}\log x - \dfrac{x^2}{4} + C$
14	$y = \cos x + 2e^{-x}$

Level 3

15	$(x-y)^2 = Cx\,e^{-y/x}$
16	$xy = e^x\left(x^2 - 2x + 2\right) - e$
17	$2e^{x/y} + \log\lvert y\rvert = 2$
18	$\dfrac{y}{x} - \log\lvert y\rvert = C$
19	$x = y^2\left(e^{-1} - e^{-y}\right)$
20	(i) $\dfrac{dP}{dt} = kP$, order 1, degree 1; (ii) $P = 20000\,e^{kt}$, $k = \dfrac{1}{5}\log\dfrac{5}{4}$; (iii) $t = 10$, the year 2029